The Entropy Ratio: Quantitative Confidentiality for Trapdoor Computing

Alexander Towell*
lex@metafunctor.com

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Abstract

A cipher map is a total function on bit strings implementing a trapdoor approximation of a latent function: the untrusted machine evaluates it blindly; only the trusted machine, holding the secret, can encode inputs and decode outputs. We develop a quantitative confidentiality theory for cipher map systems. The starting point is the entropy ratio $e = H/H^*$, a normalized form of Shannon leakage borrowed from the Quantitative Information Flow literature. We connect this measure to the cipher map framework's representation-uniformity parameter δ via the Fannes–Audenaert continuity inequality, giving $e \geq 1 - \delta - h_2(\delta)/n$. This bridge makes δ the operational design target, and we characterize the explicit costs of the framework's three δ-reduction mechanisms (noise injection, multiple representations, and encoding granularity), contributing a Fisher-information dilution result for noise injection. The paper's main result is a compositional leakage theorem with a matching minimax lower bound: even when each cipher map is marginally δ-uniform, observing multiple evaluations on a shared cipher value lets the untrusted machine recover the latent joint distribution to total-variation accuracy ξ at the *information-theoretically optimal* rate $\Theta(|Y_1| \cdot |Y_2|/\xi^2)$ (matching upper bound from plug-in estimation, matching lower bound via Assouad's lemma). The compositional channel is therefore not merely exploitable but unavoidable for any choice of per-cipher-map parameters; reducing δ is necessary but not sufficient, and only system-level interventions (observation limits, joint encoding, or noise injection) reduce the adversary's effective advantage. Experimental validation on encrypted Boolean search over the 20 Newsgroups corpus confirms the theoretical predictions for FPR compounding and the encoding granularity spectrum.

Keywords. cipher maps, entropy ratio, Fannes–Audenaert continuity, compositional leakage, homophonic substitution, quantitative information flow, trapdoor computing.

1 Introduction

Given a cipher map system—a trusted machine encoding queries, an untrusted machine evaluating total functions on bit strings, and results decoded only by the trusted machine—how much does the untrusted machine learn?

The cipher map framework [15] provides four properties (totality, representation uniformity, correctness, composability) parameterized by $(\eta, \varepsilon, \mu, \delta)$ that characterize what the untrusted machine can and cannot do. Totality ensures every input produces output; representation uniformity bounds frequency analysis; correctness governs error rates; composability enables chained evaluation. But

*ORCID: 0000-0001-6443-9897.

these properties are *qualitative* guarantees. A system designer needs *quantitative* answers: given a specific deployment with a known query distribution, known vocabulary size, and known resource budget, how confidential is the system, and how can it be improved?

Our central observation is that confidentiality in cipher map systems decomposes into two scales. At the *marginal scale*, a single cipher map's leakage is controlled by its representation-uniformity parameter δ, and the per-cipher-value entropy ratio gives a clean translation. At the *compositional scale*, observations of multiple cipher map evaluations on a shared cipher value leak the joint over latent outputs at parametric rate, independent of δ. The two scales do not reduce to each other: engineering at one does not engineer at the other. This is the structural reason cipher-map confidentiality cannot be addressed by a single number, and is the spine of the paper.

Marginal scale. The central measure is the *entropy ratio* $e = H/H^*$: the ratio of observed entropy in the cipher map output stream to the maximum entropy achievable under system constraints. The measure itself is not new; it is a normalized form of Shannon leakage studied in the Quantitative Information Flow literature (§2). The cipher map contribution is the connection: we prove that the representation-uniformity parameter δ directly lower-bounds the entropy ratio via the Fannes–Audenaert continuity inequality [7, 2],

$$e \ \geq \ 1 - \delta - h_2(\delta)/n,$$

where h_2 is the binary entropy. This reduces marginal-scale engineering to the concrete problem of minimizing δ.

The cipher map framework provides three mechanisms that reduce δ. The cipher maps paper names them and explicitly defers their cost analysis to this companion work [15, Rem. 5.1]; the *constructions* are the framework's, and our contribution at the marginal scale is their analysis as cost-attached levers (the Fisher-information dilution of the first is the one new mechanism-level result):

1. **Noise injection.** The trusted machine mixes R filler queries per N real queries; by totality the untrusted machine cannot distinguish real from filler. Cost: bandwidth $(1 + R/N)$. Effect: Fisher information for any correlation estimator is reduced by $\rho^2 = (N/(N + R))^2$ (Theorem 5.1, new here).

2. **Multiple representations.** Each element receives $K(x) \geq 1$ encodings, with $K(x) \propto D(x)$ to flatten the cipher value distribution (classical homophonic substitution: frequent elements get more code symbols so that per-cipher frequency equalizes). The flattening identity is the framework's homophonic allocation result [15, Prop. 4.1]; Theorem 5.2 attaches the cost (space: $\sum_x K(x)$ slots, with $\delta \to 0$ as the budget grows).

3. **Encoding granularity**, controlled by an entanglement parameter p that interpolates between component-wise encoding (correlation leakage) and joint encoding (space $O(|Y|^k)$), inherited from [15, Sec. 8].

Compositional scale. The constructions above all act at the marginal scale: each cipher map on its own. The paper's main result is that this is not enough, and in a strong sense. When the untrusted machine observes multiple cipher map evaluations on a shared cipher value, the joint distribution over latent outputs is recoverable at the information-theoretically optimal rate $\Theta(|Y_1| \cdot |Y_2|/\xi^2)$ (matching upper bound from plug-in estimation in Theorem 6.1; matching lower bound via Assouad's lemma in Theorem 6.2); mutual information is preserved exactly, $I(\hat{f}_1(C); \hat{f}_2(C)) =$

$I(f_1(X); f_2(X))$. No per-cipher-map parameter (including arbitrarily small δ) changes the rate. The compositional channel is intrinsic to the framework's composability, not a bug: the same property that lets the untrusted machine chain evaluations blindly is the property that creates the channel. Defending against a distribution-estimation adversary at the compositional scale therefore requires system-level interventions (reducing shared-c observations, joint encoding via Prop. 5.3, or noise injection); marginal-scale parameter tuning alone does not. We also include, for completeness, FPR compounding through Boolean chains, which is inherited from [15, Sec. 7.4].

Practical measurement. We provide practical measurement tools that close the loop between theory and operation. The entropy of a cipher map output stream can be estimated by compressing it: more compressible means more predictable means less confidential. This avoids the need for explicit probabilistic models of the adversary.

Contributions.

1. **Compositional leakage at the optimal rate (main result).** Theorems 6.1 and 6.2 together establish that when the untrusted machine observes multiple cipher map evaluations on a shared cipher value, the joint distribution over latent outputs is recoverable at the information-theoretically optimal rate $\Theta(|Y_1| \cdot |Y_2|/\xi^2)$, with mutual-information preservation $I(\hat{f}_1(C); \hat{f}_2(C)) = I(f_1(X); f_2(X))$. The upper bound is attained by the empirical-distribution plug-in estimator; the matching minimax lower bound is via Assouad's lemma. No per-cipher-map parameter (including arbitrarily small δ) changes the rate, so reducing δ is necessary but not sufficient: only system-level interventions (observation limits, joint encoding via Prop. 5.3, or noise injection) reduce the adversary's effective advantage at this scale. FPR compounding through Boolean chains is included for context; it is inherited from [15] (§6).

2. **Fannes bridge.** The representation-uniformity parameter δ directly lower-bounds the entropy ratio, $e \geq 1 - \delta - h_2(\delta)/n$, via the Fannes–Audenaert continuity inequality (Theorem 4.1). This makes δ the operational handle for confidentiality engineering in cipher map systems (§4).

3. **Cost analysis of the framework's δ-reduction levers.** The cipher maps framework supplies three mechanisms (noise injection, multiple representations, encoding granularity) and defers their cost analysis to this paper [15, Rem. 5.1]. We characterize each: noise injection (bandwidth cost $1 + R/N$) with a Fisher-information dilution ρ^2 that is new here (Theorem 5.1); multiple representations (space cost $\sum_x K(x)$ with $K(x) \propto D(x)$), building on the framework's homophonic identity [15, Prop. 4.1] (Theorem 5.2); and encoding granularity, used as inherited (§5).

4. **Practical measurement and validation.** Compression-based entropy estimation avoiding explicit adversary models, and a case study demonstrating confidentiality improvement from 72% to 98% with approximately $1.04\times$ space and $1.5\times$ bandwidth overhead (§7, §8).

What this paper does not do. We do not use ORAM, FHE, or simulation-based security definitions. Privacy in the cipher map model comes from the one-way hash, the totality of $\hat{f}$, and representation uniformity—not from access-pattern indistinguishability or computational hardness assumptions beyond the hash function. The framework is information-theoretic throughout.

2 Related Work

Quantitative information flow. The entropy ratio $e = H/H^*$ is a normalized form of Shannon leakage, studied extensively in the QIF literature [13, 1]. Smith [13] proposes min-entropy leakage as a worst-case measure; Alvim et al. [1] provide a comprehensive treatment of information-theoretic leakage measures. The measure itself is not our contribution, and we do not claim novelty there. What is new here is (i) the compositional leakage result showing that δ-uniform marginals are insufficient when shared cipher values recur across evaluations (Theorems 6.1 and 6.2) establishing the information-theoretically optimal rate $\Theta(|Y_1|\cdot|Y_2|/\xi^2)$ via matching upper and lower bounds, the paper's main result, and (ii) the *Fannes bridge* connecting the representation-uniformity parameter δ of cipher maps to the entropy ratio ($e \geq 1 - \delta - h_2(\delta)/n$), making δ the operational handle QIF does not have. The QIF literature treats leakage of a function; the cipher map setting adds a four-parameter construction $(\eta, \varepsilon, \mu, \delta)$ and a composition discipline under which that leakage must be reasoned about. We discuss the Shannon-vs-min-entropy choice in §9.

SSE leakage mitigation. Noise injection for encrypted search has been studied in the SSE literature. Bost and Fouque [3] propose dummy queries with game-based security analysis; Demertzis et al. [5] develop tunable leakage-functionality trade-offs. Our analysis differs in using information-theoretic measures (entropy ratio) rather than simulation-based definitions, and applies to general cipher maps, not only encrypted search.

Leakage from property-preserving encryption. Naveed et al. [10] and Islam et al. [8] demonstrate inference attacks exploiting preserved algebraic structure. The cipher map model avoids property preservation entirely: the untrusted machine evaluates a total function on opaque bit strings and cannot determine the output type from the cipher value.

3 Preliminaries

We recall the cipher map abstraction from [15]. All notation follows that paper; we cite rather than re-derive.

Definition 3.1 (Cipher map [15, Def. 3.1])**.** A *cipher map* for a latent function $f : X \to Y$ is a tuple $(\hat{f}, \mathrm{enc}, \mathrm{dec}, s)$ where $\hat{f} : \{0,1\}^n \to \{0,1\}^n$ is a total function on n-bit strings, $\mathrm{enc} : X \times \{0, \ldots, K(x)-1\} \to \{0,1\}^n$ is an encoding function with $K(x) \geq 1$ representations per element, $\mathrm{dec} : \{0,1\}^n \to Y \cup \{\perp\}$ is a decoding function, and s is a secret (the trapdoor). The untrusted machine holds $\hat{f}$. The trusted machine holds enc, dec, and s.

The four properties, parameterized by $(\eta, \varepsilon, \mu, \delta)$ [15, Sec. 4]:

Totality. $\hat{f}$ is defined on all 2^n inputs. Out-of-domain outputs are indistinguishable from uniform under the random oracle model.

Representation uniformity (δ-bounded). The marginal distribution of cipher values is δ-close to uniform over $\{0,1\}^n$ in total variation distance: $d_{\mathrm{TV}}(P, U) = \frac{1}{2}\sum_c |P(c) - U(c)| \leq \delta$. Informally, δ bounds the advantage of any test that tries to distinguish real cipher values from random ones.

Correctness (η-bounded). The fraction of domain elements that decode incorrectly is at most η:

$$\Pr[\mathrm{dec}(\hat{f}(\mathrm{enc}(x, k))) \neq f(x)] \leq \eta.$$

Nonzero η provides plausible deniability (an observed result might be wrong) and reduces construction cost.

Noise-decode probability (ε). A random n-bit string decodes to a valid output with probability ε. Lower ε means out-of-domain queries are more likely to be rejected (decoded as $\perp$), at the cost of more bits per element.

Value cost ($\mu = H(Y)$). The average number of bits needed to encode one output value, equal to the Shannon entropy of the output distribution.

Composability. The composition of two cipher maps is again a cipher map: $\hat{g} \circ \hat{f}$ has correctness $\eta_{g \circ f} \leq 1 - (1 - \eta_f)(1 - \eta_g)$. Errors accumulate predictably, enabling pipeline analysis.

The space per element decomposes as bits/element $= -\log_2 \varepsilon + H(Y)$, where $-\log_2 \varepsilon$ bits distinguish signal from noise and $H(Y) = \mu$ bits encode the function value [15, Thm. 6.1].

Trusted/untrusted model. The trusted machine T encodes inputs, decodes results, and injects filler queries. The untrusted machine U evaluates $\hat{f}$ on any bit string and returns results. U cannot decode, distinguish real from filler queries, determine the domain X, or enumerate which inputs are "in-domain" [15, Sec. 5].

Acceptance predicates. The batch construction of a cipher map is unified under an *acceptance predicate*: a family of disjoint sets $\{A(y) \subseteq \{0,1\}^n : y \in Y\}$ that partitions hash space among output values [15, Sec. 6.2]. Shannon-optimal allocation sets $|A(y)| \propto \Pr[f(X) = y]$, simultaneously minimizing space and maximizing output indistinguishability.

Information-theoretic notation. We use standard information-theoretic quantities throughout. *Shannon entropy* $H(X) = -\sum_x p(x) \log_2 p(x)$ measures the average surprise (in bits) of a random variable; higher entropy means less predictable. *Conditional entropy* $H(X \mid Y)$ is the remaining uncertainty in X after observing Y. *Mutual information* $I(X;Y) = H(X) - H(X \mid Y)$ quantifies how much observing Y reveals about X; zero means independence. *KL divergence* $D_{\mathrm{KL}}(P\|Q) = \sum_x P(x) \log_2(P(x)/Q(x))$ measures how far P is from Q; it is zero iff $P = Q$ and connects to the entropy ratio via $H(P) = H^* - D_{\mathrm{KL}}(P\|U)$ when U is the maximum-entropy (uniform) distribution. We use D for the distribution of queries over the domain X, and Q for the induced distribution over cipher values. All logarithms are base 2.

4 The Confidentiality Measure

This section and §5 develop the marginal-scale theory introduced in §1: how to measure single-cipher-map leakage via the entropy ratio, and how to engineer toward the design target δ. The compositional-scale theory is in §6.

4.1 Observed and Maximum Entropy

Consider a cipher map system in which the trusted machine submits a stream of cipher values to the untrusted machine. The untrusted machine observes a sequence $c_1, c_2, \ldots, c_N$ of n-bit strings, each either a real encoding $\mathrm{enc}(x_i, k_i)$ or a filler string drawn uniformly from $\{0,1\}^n$.

Definition 4.1 (Observed entropy)**.** The *observed entropy* H of a cipher map system is the Shannon entropy of the joint distribution over the observable sequence:

$$H = H(C_1, C_2, \ldots, C_N), \tag{1}$$

where each C_i is the random variable corresponding to the i-th cipher value observed by the untrusted machine.

Definition 4.2 (Maximum entropy under constraints)**.** The *maximum entropy* H^* is the maximum of H over all distributions on the observable sequence that are consistent with the system constraints: domain size $|X|$, codomain size $|Y|$, and the cipher map parameters $(\eta, \varepsilon, \delta)$.

By the maximum entropy principle [9], H^* is achieved when the observable distributions are as uniform as possible subject to the constraints. For a single cipher value drawn from a vocabulary of size m, the maximum entropy is $\log_2 m$ (uniform distribution). For a sequence of N independent cipher values, the maximum entropy is $N \log_2 m$.

4.2 The Entropy Ratio

Definition 4.3 (Entropy ratio)**.** The *entropy ratio* of a cipher map system is

$$e = \frac{H}{H^*} \in [0, 1]. \tag{2}$$

The entropy ratio has a direct operational interpretation:

- $e = 0$: the output sequence is fully predictable. The untrusted machine can predict every cipher value with certainty. No confidentiality.
- $e = 1$: the output sequence is indistinguishable from the maximum entropy distribution under system constraints. The untrusted machine's best prediction is no better than guessing from the maximum entropy distribution. Maximum confidentiality.

4.3 Relationship to Cipher Map Parameters

The entropy ratio connects to the four cipher map parameters through the following result.

Theorem 4.1 (Entropy ratio decomposition)**.** *Let $(\hat{f}, \mathrm{enc}, \mathrm{dec}, s)$ be a cipher map for $f : X \to Y$ with parameters $(\eta, \varepsilon, \mu, \delta)$. Let D be the query distribution on X, and let Q be the induced distribution on cipher values. Then the per-query observed entropy satisfies:*

$$H(Q) = H^* - D_{\mathrm{KL}}(Q\|U) = H^*\left(1 - \frac{D_{\mathrm{KL}}(Q\|U)}{H^*}\right), \tag{3}$$

where U is the uniform distribution on $\{0,1\}^n$ and the entropy ratio is $e = H(Q)/H^* = 1 - D_{\mathrm{KL}}(Q\|U)/H^*$. In particular:

1. *If $\delta = 0$ (perfect representation uniformity), then $Q = U$ and $H(Q) = H^* = n$ bits, giving $e = 1$.*
2. *If $K(x) = 1$ for all x (simple substitution, no multiple representations), then Q inherits the non-uniformity of D, and $e \leq H(D)/\log_2 |X|$.*

3. *For $\delta \leq 1/2$, the entropy ratio is bounded by*

$$e \geq 1 - \delta - h_2(\delta)/n,$$

where $h_2(\delta) = -\delta \log_2 \delta - (1-\delta)\log_2(1-\delta)$ is the binary entropy. The bound follows from the Fannes–Audenaert continuity inequality [7, 2].

Proof. For part (1): when $\delta = 0$, Q is uniform over $\{0,1\}^n$ by definition of representation uniformity. The entropy of the uniform distribution on 2^n elements is n bits, which equals H^*.

For part (2): with $K(x) = 1$, the encoding enc is injective, so Q is a relabeling of D. Entropy is invariant under bijection, giving $H(Q) = H(D)$. The maximum over distributions on $|X|$ elements is $\log_2 |X|$, so $e = H(D)/\log_2 |X|$.

For part (3): the Fannes–Audenaert inequality bounds entropy continuity: for any distributions P, Q on a support of size d with $d_{\mathrm{TV}}(P,Q) \leq t \leq 1/2$,

$$|H(P) - H(Q)| \;\leq\; t\log_2(d-1) + h_2(t).$$

Applying this to Q and U on $\{0,1\}^n$ with $d = 2^n$ and $d_{\mathrm{TV}}(Q,U) \leq \delta$,

$$|H(Q) - n| \;\leq\; \delta\log_2(2^n - 1) + h_2(\delta) \;\leq\; \delta n + h_2(\delta),$$

so $H(Q) \geq n(1-\delta) - h_2(\delta)$ and $e = H(Q)/n \geq 1 - \delta - h_2(\delta)/n$. Note the bound is linear in δ (not quadratic): Pinsker gives $D_{\mathrm{KL}} \geq 2\, d_{\mathrm{TV}}^2$, an upper bound on TV given KL, which does not convert to a KL bound given TV. The correct direction is Fannes–Audenaert. □

Remark 4.1 (Two normalizers for H^*). The theorem uses two values of H^* for two different questions. The *ambient* entropy ratio normalizes by $H^* = n$, the entropy of the uniform distribution on the full cipher space $\{0,1\}^n$; this is the quantity the Fannes bound (part 3) controls and the canonical marginal measure (it is what "the entropy ratio" means without qualification). The *effective-alphabet* ratio in part (2) normalizes by $H^* = \log_2 |X|$, the entropy of the uniform distribution over the $|X|$ in-domain values, and answers a narrower question: how much of the *achievable* entropy a simple substitution captures. When $|X| \ll 2^n$ the two differ; the worked entropy ratios reported in the examples and case study of §5 and §8 use the effective-alphabet normalizer $\log_2 |X|$, since the cipher space is sized to the vocabulary.

Remark 4.2 (Numerical scale). Take a 64-bit cipher-value space ($n = 64$, a typical width for the hash output of a cipher map). For $\delta = 0.05$, the bound gives $e \geq 0.945$. For $\delta = 0.01$, $e \geq 0.989$. The linear scaling in δ means tightening δ by an order of magnitude tightens the leakage bound by roughly the same order.

Remark 4.3 (The role of ε). The noise decode probability ε does not appear directly in the per-query entropy, but it determines the space cost: achieving small ε requires $-\log_2 \varepsilon$ bits per element. The indirect effect is that larger ε means a denser cipher space (more random bit strings decode to valid outputs), which helps uniformity but increases false positives.

Remark 4.4 (The role of η). The correctness parameter η provides a form of plausible deniability. With $\eta > 0$, the untrusted machine cannot be certain that a decoded output reflects the true function value. This adds $H_b(\eta) = -\eta\log_2 \eta - (1-\eta)\log_2(1-\eta)$ bits of uncertainty per query (the binary entropy of the error event).

5 Constructions for Reducing δ

The cipher map framework supplies three mechanisms that reduce δ (§1); this section analyzes each as a cost-attached lever. The constructions are the framework's; the cost characterizations, and the Fisher-information dilution of noise injection (§5.1), are this paper's marginal-scale contribution.

5.1 Noise Injection

The trusted machine can mix R filler queries with N real queries. By totality, the untrusted machine cannot distinguish real from filler: both are n-bit strings, and $\hat{f}$ produces n-bit output on both.

Theorem 5.1 (Noise dilution)**.** *Assume $K(x) = 1$ (single encoding per element), so the cipher value distribution of real queries equals D. Let the trusted machine submit N real queries drawn from D and R filler queries drawn uniformly from $\{0,1\}^n$. The mixed sequence has $N+R$ elements, each of which is real with probability $\rho = N/(N+R)$ and filler with probability $1-\rho$. Then:*

1. *The per-element entropy of the mixed distribution is*

$$H_{\text{mix}} = H_b(\rho) + \rho \cdot H(D) + (1-\rho) \cdot n, \tag{4}$$

where $H_b(\rho)$ is the binary entropy of ρ.

2. *As $R/N \to \infty$, $H_{\text{mix}} \to n = H^*$ and $e \to 1$.*

3. *For any one-dimensional parameter θ of D, the Fisher information satisfies*

$$I^{\text{mix}}(\theta) \;\leq\; \rho^2 \cdot C(D) \cdot I^{\text{pure}}(\theta), \tag{5}$$

where $I^{\text{pure}}(\theta)$ is the Fisher information from observing D directly and $C(D) = \max_c D(c)/U(c) = 2^n \max_c D(c)$ is a distribution-dependent constant capturing the worst-case concentration of D relative to uniform. In particular, for $\rho \ll 1$ the variance of any unbiased estimator scales as $\Omega\big((1+R/N)^2/C(D)\big) \cdot \mathrm{Var}_{\text{pure}}$.

Proof. For part (1): each observed element is a mixture. With probability ρ it is drawn from D (entropy $H(D)$); with probability $1-\rho$ it is uniform on $\{0,1\}^n$ (entropy n). By the chain rule, the per-element entropy is the entropy of the mixing indicator plus the conditional entropy given the indicator.

For part (2): as $\rho \to 0$, the distribution converges to uniform on $\{0,1\}^n$.

For part (3): write the mixture density at observation c as $P_{\text{mix}}(c \mid \theta) = \rho P_\theta(c) + (1-\rho)U(c)$, where $U(c) = 2^{-n}$ is uniform. The score is $\partial_\theta \log P_{\text{mix}} = \rho \partial_\theta P_\theta / P_{\text{mix}}$, so

$$I^{\text{mix}}(\theta) = \mathbb{E}_{\text{mix}}\Big[\big(\partial_\theta \log P_{\text{mix}}\big)^2\Big] = \rho^2 \int \frac{(\partial_\theta P_\theta)^2}{P_{\text{mix}}}\, dc.$$

Bounding $P_{\text{mix}}(c) \geq (1-\rho)U(c) = (1-\rho)2^{-n}$ in the denominator and $P_{\text{mix}}(c) \leq \rho P_\theta(c) + U(c) \leq \max\{P_\theta(c), U(c)\} \cdot 2$ gives $I^{\text{mix}}(\theta) \leq \rho^2 \cdot C(D) \cdot \int (\partial_\theta P_\theta)^2 / P_\theta \, dc = \rho^2 \cdot C(D) \cdot I^{\text{pure}}(\theta)$, where the constant $C(D) = \max_c P_\theta(c)/U(c) = 2^n \max_c P_\theta(c)$ absorbs the worst-case concentration of D. The Cramér-Rao bound then gives the variance scaling claim. Note that $C(D)$ is unbounded as D becomes a point mass; the ρ^2 scaling is exact only for distributions with bounded ratio to uniform. □

Cost. Noise injection costs bandwidth: the total query volume increases by a factor of $(N+R)/N = 1 + R/N$. It does not cost space (no additional cipher map storage). A noise ratio of $R/N = 1$ (equal filler and real queries) doubles bandwidth; $R/N = 9$ (90% filler) increases bandwidth tenfold but brings e close to 1 for most practical distributions.

5.2 Multiple Representations ($K > 1$)

Each element x can be given $K(x) \geq 1$ distinct encodings. The trusted machine, when encoding x, chooses k uniformly from $\{0, \ldots, K(x)-1\}$, so the probability of observing a particular cipher value $v = \mathrm{enc}(x, k)$ is $D(x)/K(x)$ (assuming distinct encodings, which holds with high probability under the random oracle model). Flattening the cipher value distribution therefore requires $D(x)/K(x)$ to be constant in x: more frequent elements need *more* representations, not fewer. This is the classical homophonic substitution prescription [12]: $K(x) \propto D(x)$.

Theorem 5.2 (Representation uniformity via multiplicity)**.** *Let D be a distribution on X with $D(x) > 0$ for all x, and let $c \geq 1/\min_x D(x)$ be a normalizing constant. Set $K(x) = \lceil c \cdot D(x) \rceil$, and write $N = \sum_x K(x)$. Then the cipher value distribution Q induced on the image of* enc *satisfies the homophonic allocation identity [15, Prop. 4.1],*

$$d_{\mathrm{TV}}(Q, U_{\mathrm{im}}) \;=\; d_{\mathrm{TV}}\big(D,\, K/N\big) \;<\; \frac{|X|-1}{N} \;\leq\; \frac{|X|-1}{c}, \tag{6}$$

where U_{im} is the uniform distribution on $\mathrm{im(enc)}$ *and the constant $|X|-1$ is tight. In particular, $d_{\mathrm{TV}}(Q, U_{\mathrm{im}}) \to 0$ as $c \to \infty$. Under the random oracle model,* $\mathrm{im(enc)}$ *is a pseudorandom subset of $\{0,1\}^n$, so $d_{\mathrm{TV}}(Q, U)$ inherits the same bound up to a birthday-bound term $O(N/2^n)$.*

Proof. Under the random oracle model, distinct (x, k) pairs map to distinct cipher values with high probability, so enc is injective on the image and $Q(v) = D(x)/K(x)$ for the unique (x, k) with $\mathrm{enc}(x, k) = v$. The homophonic allocation proposition [15, Prop. 4.1] gives the exact identity $d_{\mathrm{TV}}(Q, U_{\mathrm{im}}) = d_{\mathrm{TV}}(D, K/N)$ and bounds the Simmons allocation $K(x) = \lceil c\, D(x) \rceil$ by $(|X|-1)/N$, with the constant $|X|-1$ tight (mass concentrated on one value, the remaining $|X|-1$ values sitting just above the rounding boundary, attains it). This paper's addition is the cost reading: the bound is bought with $N = \sum_x K(x)$ cipher-map slots, so driving δ down trades linearly against space. The image-to-$\{0,1\}^n$ step follows from the random-oracle assumption: a pseudorandom image of size N inside $\{0,1\}^n$ contributes at most $O(N/2^n)$ additional TV distance. □

Cost. Multiple representations cost space: each element now occupies $K(x)$ slots in the cipher map, so the total space is $\sum_x K(x) \cdot (-\log_2 \varepsilon + H(Y))$ bits. With $K(x) = \lceil c \cdot D(x) \rceil$, the total number of representations is $\sum_x K(x) \approx c$ (plus a rounding term bounded by $|X|$). The cost scales with the normalizing constant c, which sets the achievable δ via Theorem 5.2.

Example 5.1 (Homophonic encryption for Zipf-distributed queries)*.* A vocabulary of m = 10,000 words with Zipf distribution $D(x_i) \propto 1/i$ has harmonic normalizer $H_m = \sum_{i=1}^{m} 1/i \approx 9.788$, entropy $H(D) \approx 9.55$ bits, and $H^* = \log_2 10{,}000 \approx 13.29$ bits, giving baseline $e \approx 0.72$. Flattening the top $b = 100$ words via $K(x_i) = \lceil c \cdot D(x_i) \rceil$ with $c = 100 \cdot H_m \approx 979$ yields $K(x_1) \approx 100, K(x_2) \approx 50, \ldots, K(x_{100}) \approx 1$, total $\sum_{i=1}^{100} K(x_i) \approx 100 \cdot H_{100} \approx 519$ cipher cells for the top words. Each of these cells has mass $D(x_i)/K(x_i) \approx 1/979$, so the top-100 mass ≈ 0.53 is spread approximately uniformly across 519 cells. The effective entropy rises to $H(Q) \approx 11.5$ bits, giving $e \approx 0.87$. The space overhead is approximately 519 additional trapdoors, a $1.04\times$ increase over the baseline 10,000 trapdoors.

5.3 Encoding Granularity

Representation uniformity is a marginal property: it says the cipher value distribution for each individual element is close to uniform. It says nothing about the joint distribution of multiple cipher values. The *encoding granularity* controls what correlations are hidden.

Definition 5.1 (Entanglement parameter [15, Sec. 8])**.** For a system encoding k correlated values, the *entanglement parameter* p is the number of values encoded as a single cipher map unit. The system has type $\text{cipher}(\{0,1\}^p)^{k/p}$, where each block of p values is encoded jointly.

Proposition 5.3 (Granularity spectrum [15, Prop. 8.1])**.** *For a cipher map encoding a pair of values $(a, b) \in A \times B$:*

1. ***Joint encoding*** *($p = 2$): the pair (a, b) is encoded as a single cipher value. The untrusted machine cannot distinguish (a_1, b_1) from (a_2, b_2) (up to δ), but projections require cipher map constructions by the trusted machine.*

2. ***Component-wise encoding*** *($p = 1$): a and b are encoded independently. Projections are free, but the joint distribution of (a, b) is preserved in the cipher pair. With N observations, the adversary estimates the joint distribution to accuracy ξ in TV distance from $O(|A| \cdot |B|/\xi^2)$ samples.*

The encoding granularity creates a spectrum of confidentiality/ functionality trade-offs, analyzed in detail in [14]:

Granularity	Intermediates	Correlation hiding	Space	Projections
Root ($p = k$)	0	Full	$O(\lvert Y\rvert^k)$	None free
Intermediate	few	Partial	moderate	Some free
Leaf ($p = 1$)	k	None (marginal only)	$O(k \cdot \lvert Y\rvert)$	All free

The sum-type impossibility theorem [14, Thm. 4.2] shows that this trade-off is fundamental for sum types: for $A + B$, tag hiding and untrusted pattern matching are mutually exclusive. Products leak correlations; sums leak the tag.

Confidentiality bound from orbit closure. The orbit closure [14, Sec. 5] quantifies the dynamic confidentiality cost of exposing operations. Given cipher maps $F = \{\hat{f}_1, \ldots, \hat{f}_m\}$ available to the untrusted machine and a starting cipher value c, the *orbit* $\text{orbit}_F(c)$ is the set of all cipher values reachable from c by composing operations in F. The *confidentiality* $\text{conf}_F(c)$ measures how much uncertainty the adversary retains about the latent value encoded by c: it is 1 when the adversary has learned nothing and 0 when the latent value is uniquely determined. The orbit size bounds it. Theorem 5.3 of [14] gives the entropy form $H(X \mid \mathcal{V}_F(c)) \geq H(X) - \log_2 |\text{orbit}_F(c)|$; the normalized set form (over the latent domain X, not the ambient cipher space) is

$$\text{conf}_F(c) \geq 1 - \frac{|\text{orbit}_F(c)|}{|X|}. \tag{7}$$

A larger orbit (more reachable cipher values) means the adversary has explored more of the latent space, leaving fewer possibilities for the latent value. Fewer exposed operations means smaller orbit means higher confidentiality. This provides a formal justification for the root encoding strategy: exposing no intermediate cipher values gives $|\text{orbit}_\emptyset(c)| = 1$ and confidentiality $\geq 1 - 1/|X|$.

6 The Compositional Scale

When cipher maps compose, as in $\hat{g} \circ \hat{f}$ or Boolean combinations like $\widehat{\text{AND}}(\hat{f}(c), \hat{g}(c))$, confidentiality moves to a different scale than the marginal one addressed in §5. Composition has both a *confidentiality* facet and a *correctness* facet, and we treat both in this section. Section 6.1 covers the confidentiality facet, the compositional scale proper: correlation leakage from shared cipher values, the paper's headline result, exposing a fundamental limit of the marginal-scale constructions of §5 (reducing δ to zero is necessary but not sufficient when cipher values recur). Section 6.2 covers the correctness facet: how the cipher map error parameter η compounds through Boolean chains. This is inherited from [15, Sec. 7.4] and included here both for completeness and because it interacts with the confidentiality story (e.g., AND chains drive false-positive *noise* down, which has confidentiality consequences treated in §6.2).

6.1 Correlation Leakage from Shared Variables

Even when each cipher map has uniform marginal output, evaluating multiple cipher maps on the same cipher value leaks correlations [15, Sec. 8.2]. If the untrusted machine evaluates $\hat{f}_1(c)$ and $\hat{f}_2(c)$ for the same c, the pair $(\hat{f}_1(c), \hat{f}_2(c))$ is a deterministic function of c, and the joint distribution preserves the latent correlation between f_1 and f_2. This is the central compositional result of the paper.

Theorem 6.1 (Compositional leakage). *Let $\hat{f}_1, \hat{f}_2$ be cipher maps for latent $f_1 : X \to Y_1$, $f_2 : X \to Y_2$, each with $\delta_i \approx 0$ so their marginal cipher value distributions are near-uniform on $\{0,1\}^n$. Assume the untrusted machine observes pairs $(\hat{f}_1(c_i), \hat{f}_2(c_i))$ for $i = 1, \ldots, N$, where each c_i is an independent in-domain cipher value drawn according to the pushforward of D under* enc *(so the latent pair $(f_1(x), f_2(x))$ is drawn i.i.d. from the true joint under D). Then:*

1. *Each marginal cipher output is individually δ_i-uniform on $\{0,1\}^n$ (by hypothesis).*
2. *Mutual information is preserved: $I(\hat{f}_1(C); \hat{f}_2(C)) = I(f_1(X); f_2(X))$, where $X \sim D$ and $C = \mathrm{enc}(X, k)$ for uniform k.*
3. *The adversary estimates the joint distribution on $Y_1 \times Y_2$ to TV accuracy ξ from $N = O(|Y_1| \cdot |Y_2|/\xi^2)$ samples, the standard plug-in estimator rate.*
4. *Any downstream cipher map $\hat{f}_3$ with Shannon-optimal acceptance predicate for the* marginal *output distribution remains non-uniform when applied to the correlated pair: achieving $\delta_3 \approx 0$ requires reconstructing $\hat{f}_3$ with respect to the joint distribution.*

Proof. Part (1) is the hypothesis. Part (2): the maps $\mathrm{dec}_i \circ \hat{f}_i$ push cipher pairs back to latent pairs deterministically (up to η). Under the sampling assumption, the encoding randomness k is drawn uniformly from $\{0, \ldots, K(X) - 1\}$ marginally independent of the latent pair $(f_1(X), f_2(X))$, so the observed joint and the latent joint have equal mutual information. Data-processing then gives equality because the push maps are surjective onto Y_1, Y_2 on the in-domain part. Part (3): the empirical distribution on a discrete support of size $|Y_1| \cdot |Y_2|$ converges in TV to the true distribution at rate $O(\sqrt{|Y_1| \cdot |Y_2|/N})$ [4], so $N = O(|Y_1| \cdot |Y_2|/\xi^2)$ samples suffice for TV accuracy ξ. Part (4): when $I(f_1; f_2) > 0$, the joint input distribution to $\hat{f}_3$ is not uniform on $Y_1 \times Y_2$, so a predicate optimized for the product of marginals cannot achieve $\delta_3 = 0$; the optimal δ_3 is bounded below by the TV distance between the joint and the product of marginals. □

The upper bound of Theorem 6.1 part (3) is matched by a minimax lower bound: no estimator can recover the joint distribution at a faster rate than the empirical distribution does.

Theorem 6.2 (Compositional leakage is rate-optimal)**.** *Under the sampling model of Theorem 6.1, any estimator $\hat{P}$ of the joint distribution P on $Y_1 \times Y_2$ based on N i.i.d. samples of $(\hat{f}_1(C_i), \hat{f}_2(C_i))$ satisfies*

$$\inf_{\hat{P}} \sup_{P \in \Delta(Y_1 \times Y_2)} \mathbb{E}\big[d_{\mathrm{TV}}(\hat{P}, P)\big] \;\geq\; c\sqrt{|Y_1| \cdot |Y_2|/N}, \tag{8}$$

for an absolute constant $c > 0$, where the supremum is over joint distributions on $Y_1 \times Y_2$ with full support. Achieving expected TV accuracy ξ therefore requires $N = \Omega(|Y_1| \cdot |Y_2|/\xi^2)$ samples in the worst case, matching the upper bound of Theorem 6.1 part (3) up to constants. The minimax rate for compositional joint recovery is $\Theta(|Y_1| \cdot |Y_2|/\xi^2)$.

Proof sketch. This is the standard Assouad minimax bound for distribution estimation in TV, applied to the joint support $Y_1 \times Y_2$.[1] Let $m = |Y_1| \cdot |Y_2|$ and pair the cells of $Y_1 \times Y_2$ into $m/2$ disjoint pairs $\{(v_i^+, v_i^-)\}_{i=1}^{m/2}$. For $s \in \{\pm 1\}^{m/2}$ and ε a small constant times the target ξ, define $P^{(s)}$ by

$$P^{(s)}(v_i^{\pm}) = \tfrac{1}{m}(1 \pm s_i \varepsilon),$$

so each $P^{(s)}$ is a perturbation of uniform on $Y_1 \times Y_2$ with total mass 1. Pairwise TV is $d_{\mathrm{TV}}(P^{(s)}, P^{(s')}) = (2\varepsilon/m)\, d_H(s, s')$ where d_H is Hamming distance (each differing coordinate moves mass ε/m at v_i^+ and ε/m at v_i^-, contributing $2\varepsilon/m$ to TV). Assouad's lemma applied to this $2^{m/2}$-packing, with KL divergence between adjacent hypotheses bounded by $O(\varepsilon^2/m)$ per coordinate (a standard consequence of the bounded-perturbation construction), gives the minimax rate $\Omega(\sqrt{m/N}) = \Omega(\sqrt{|Y_1| \cdot |Y_2|/N})$ in expected TV. See Devroye, Györfi, and Lugosi [6] for the canonical treatment of the construction and constants. □

Remark 6.1 (Sampling regimes). The bounds in Theorems 6.1 and 6.2 both assume each observation uses an independent c_i drawn from the same distribution. A different regime, in which the same cipher value c is reused across evaluations, does not yield additional samples of the joint and so has no corresponding rate. The interesting case for compositional leakage is many distinct c_i observed across deployment, each evaluated under both $\hat{f}_1$ and $\hat{f}_2$.

Why this matters. The constructions in §5 all target the marginal parameter δ. Theorem 6.1 shows that the adversary *can* recover the joint at the rate $O(|Y_1| \cdot |Y_2|/\xi^2)$ when $\delta \to 0$; Theorem 6.2 shows that no estimator can do better. The compositional channel is therefore not merely exploitable, it is information-theoretically minimax-optimal for the adversary: the cipher map system designer cannot push the adversary's task to a harder rate by any choice within the marginal-scale tools. The only defenses are reducing the number of shared-c observations ($N \lesssim |Y_1| \cdot |Y_2|/\xi^2$), reducing granularity by joint encoding (Prop. 5.3), or injecting noise (§5.1).

Mitigation strategies. Two approaches, both costly [15, Sec. 8.2]:

Build the composition directly. Construct a single cipher map for $g(x) = f_3(f_1(x), f_2(x))$. The untrusted machine sees only input and final output; no intermediates are exposed. The domain is $|X|$ (same as either component), but the construction must evaluate f_1, f_2, and f_3 for every $x \in X$ at build time.

[1] The two-point method (Le Cam) appears at both confidentiality scales of the cipher map framework, in opposite roles. At the *marginal* scale it gives an *upper* bound on a single-guess membership attacker (accuracy $\leq \frac{1}{2} + \frac{\delta}{2}$; see the companion cipher maps paper). Here, at the *compositional* scale, we need a *lower* bound on a dimension-dependent estimation rate, which a two-hypothesis argument cannot deliver; the hypercube packing below is Assouad's lemma.

A partial merge is also possible: merge f_1 and f_2 into a tuple cipher map $\widehat{(f_1, f_2)}(c) \to \mathsf{C}(Y_1 \times Y_2)$, then build $\hat{f}_3$ over the product codomain. This hides the correlation between f_1 and f_2 but the codomain $|Y_1| \times |Y_2|$ grows as a product of the component codomains. For k merged components, the codomain is $\prod_i |Y_i|$, which grows exponentially. This is the fundamental space cost of the encoding granularity principle: hiding correlations requires encoding into product spaces.

Compose components and inject noise. Keep practical component-wise evaluation and add R noise queries per N real queries. Cost: bandwidth, not space. The adversary's correlation estimates are diluted by the noise fraction.

6.2 Error Compounding in Boolean Chains

For Boolean-valued cipher maps (e.g., set indicator functions $\hat{1}_A : \mathsf{C}(X) \to \mathsf{C}(\text{Bool})$, or any $\hat{f}$ with $Y = \{\text{True}, \text{False}\}$), the composition behavior depends on the gate type. Let p_T denote the *false positive rate* (FPR):

$$p_T = \Pr[\text{dec}(\hat{f}(\text{enc}(x, k))) = \text{True} \mid f(x) = \text{False}],$$

the probability that an element with $f(x) = \text{False}$ is decoded as True. For in-domain elements with $\eta = 0$, FPR is zero by construction. For out-of-domain elements (not in the construction domain), the hash output is effectively random and lands in the True region with probability $|T|/2^n$.

Proposition 6.3 (FPR compounding). *For k independent Boolean-valued cipher maps composed via Boolean operations:*

$$\textit{AND of } k \textit{ tests:} \quad \text{FPR}_{\text{total}} = p_T^k, \tag{9}$$

$$\textit{OR of } k \textit{ tests:} \quad \text{FPR}_{\text{total}} = 1 - (1 - p_T)^k. \tag{10}$$

These follow from independence of the gate inputs (elementary). The gate-type framing is from the cipher maps paper [15, Sec. 7.4]; the explicit closed forms and their empirical validation, including the construction-level noise floor discussed below, are reported in the algebraic cipher types paper [14, Table 3].

AND drives false positive rate down exponentially: each additional conjunct is an independent filter. OR drives it up: each additional disjunct adds false positive mass. NOT is approximate due to complement non-preservation via the pigeonhole principle [15].

Effect on confidentiality. AND chains improve precision (fewer false positives) but maintain the same recall (elements with $f(x) = \text{True}$ pass all tests). From a confidentiality perspective, AND chains reduce the effective noise level: with $p_T = 0.05$ and $k = 3$, the FPR drops to $0.05^3 \approx 0.000125$, meaning almost all decoded True outputs correspond to elements where $f(x) = \text{True}$. The adversary can infer with higher confidence that a True result is correct (lower entropy ratio e), a direct trade-off between precision and confidentiality.

OR chains have the opposite effect: FPR increases, adding noise to the output stream. The adversary's uncertainty grows (higher e), but the system returns more false positives, reducing its usefulness.

Convergence under deep composition. Long AND chains converge toward False (or noise). With $p_T = 0.05$ and $p_F = 0.90$, each AND step multiplies the probability of a True result by p_T for noise inputs. After k ANDs with independent random cipher values, the probability of a True output is at most p_T^k, which vanishes rapidly. The cipher Boolean output after a deep AND chain is almost certainly False or noise, regardless of the initial value. This limits the depth of useful AND composition: beyond a few terms, the signal (legitimate True results) is indistinguishable from the False/noise floor.

Dually, long OR chains converge toward True (or noise). Each OR step with an independent input produces True with probability at least p_T, and after k steps the probability of *not* seeing True is at most $(1 - p_T)^k$. Deep OR chains saturate at True, again drowning the signal.

In both cases, noise acts as an attractor: AND pulls toward False/noise, OR pulls toward True/noise. The noise region prevents the adversary from distinguishing "legitimately False after AND" from "noise-induced False," providing a floor of uncertainty even in deep compositions.

Active probing via Boolean operations. Exposing AND and OR to the untrusted machine creates an orthogonal confidentiality risk: the adversary can actively probe cipher values to learn their latent meaning. Given a cipher value c and cipher AND, the adversary computes $\text{AND}(c, c)$. Since cipher maps are deterministic, this always returns the same result for a given c. If the adversary also has cipher values c_1, c_2 and computes $\text{AND}(c_1, c_2)$, $\text{AND}(c_1, c_1)$, and $\text{AND}(c_2, c_2)$, it can check whether the results are structurally consistent with both being True, both False, or one of each. Over many such probes, the adversary partitions cipher values into equivalence classes that correspond to latent Boolean values.

This is exactly the orbit closure attack [14, Sec. 5]: each Boolean operation enlarges the orbit of reachable cipher values from c, and the confidentiality degrades as $\text{conf}_F(c) \geq 1 - |\,\text{orbit}_F(c)|/|X|$. The more operations the untrusted machine can apply, the larger the orbit, the lower the confidentiality. This creates a fundamental tension: exposing AND/OR/NOT enables untrusted-side Boolean search (a functionality gain) but simultaneously provides the adversary with tools to probe the cipher space (a confidentiality cost).

One mitigation: *typed composition chains* [14, Sec. 5.5]. Instead of a single shared cipher Boolean space where AND can be self-composed indefinitely, define a tower of distinct cipher spaces $\mathsf{C}(\text{Bool})_0, \mathsf{C}(\text{Bool})_1, \ldots$ where AND_i maps from level i to level $i + 1$. Without AND_k, the chain terminates at depth k. The orbit is bounded by $1 + k$ and the type system enforces the limit at construction time.

7 Practical Measurement

7.1 Compression-Based Entropy Estimation

The entropy of a sequence can be estimated without an explicit probabilistic model by compressing it. This follows from Shannon's source coding theorem [11]: the expected length of the output of an optimal lossless compressor equals the entropy of the source.

Proposition 7.1 (Compression estimator). *Let* $\mathbf{c} = (c_1, \ldots, c_N)$ *be a sequence of cipher values observed by the untrusted machine, encoded as a bit string via some fixed encoding* Encode. *Let* Compress *be a lossless compressor. Then:*

$$\hat{H} = \frac{|\text{Compress}(\text{Encode}(\mathbf{c}))|}{N} \tag{11}$$

is a positively biased estimator of the per-element entropy $H(C_1, \ldots, C_N)/N$. That is, $\hat{H} \geq H/N$ in expectation, with equality when Compress *is optimal.*

Proof. A uniquely decodable code on $\mathbf{c}$ of length $L(\mathbf{c}) = |\text{Compress}(\text{Encode}(\mathbf{c}))|$ satisfies the Kraft inequality and induces a probability measure $Q(\mathbf{c}) \leq 2^{-L(\mathbf{c})}$. The expected codelength satisfies $\mathbb{E}_P[L(\mathbf{c})] \geq H(C_1, \ldots, C_N)$ by the source-coding theorem [11], with the gap equal to the divergence $D_{\mathrm{KL}}(P\|Q) \geq 0$ (cross-entropy inequality, see [4, Thm. 5.4.3]). Equality holds when the compressor matches the source distribution; for universal compressors (e.g., LZ77, LZ78), the per-symbol gap vanishes asymptotically as $N \to \infty$ [11]. Dividing by N gives $\hat{H} \geq H/N$ in expectation. The encoding overhead is bounded by a constant factor (any uniquely decodable encoding has a bounded prefix), so it contributes $O(1/N)$ to the per-symbol bias and vanishes in the same limit. □

The compression estimator has three practical advantages:

1. It requires no model of the query distribution.
2. It captures all forms of structure (frequency, correlation, temporal patterns) that a lossless compressor can detect.
3. It is cheap to compute: a single pass through the data with `gzip` or `zstd`.

Entropy ratio estimator. Given the compression-based estimate $\hat{H}$ and a computable H^* (from the system parameters), the entropy ratio is estimated as:

$$\hat{e} = \frac{\hat{H}}{H^*}. \tag{12}$$

Since $\hat{H}$ is positively biased, $\hat{e}$ is also positively biased (overestimates confidentiality). This is the conservative direction: the system is at least as confidential as the estimate suggests.

7.2 Monte Carlo Estimation

When the query distribution is available (e.g., from historical logs), entropy can be estimated by sampling. Generate M independent query sequences of length N, compute the cipher value distribution, and estimate H directly:

$$\hat{H}_{\mathrm{MC}} = -\sum_{v} \hat{p}(v) \log_2 \hat{p}(v), \tag{13}$$

where $\hat{p}(v)$ is the empirical frequency of cipher value v across the M samples. This estimator is negatively biased for finite M (the Miller–Madow correction adds $(\hat{m}-1)/(2M \ln 2)$ where $\hat{m}$ is the number of distinct observed values), but converges to the true entropy as $M \to \infty$.

7.3 The Leakage Analyzer

We define the *leakage analyzer* as the procedure that combines compression-based and Monte Carlo estimation:

1. Observe or simulate a sequence of cipher values.
2. Compute $\hat{H}$ via compression and $\hat{H}_{\mathrm{MC}}$ via sampling (if the distribution is available).
3. Compute H^* from the system parameters using the maximum entropy formulas.

4. Report $\hat{e} = \hat{H}/H^*$ and the entropy gap $H^* - \hat{H}$ in bits.

The entropy gap quantifies how many bits of information per query the adversary potentially gains from the non-uniformity of the system.

7.4 Finite-Sample Resolution of δ

The estimators above measure entropy. The representation-uniformity parameter $\delta = d_{\mathrm{TV}}(Q, U)$ can also be measured directly from observed cipher values, but only at a query cost that scales with the allocated budget.

Let $|\mathrm{Im}| = \sum_x K(x)$ denote the image size of enc under a given homophonic allocation, and let $\hat{Q}_N$ be the empirical distribution over the image after observing N cipher values. By Glivenko–Cantelli, $d_{\mathrm{TV}}(\hat{Q}_N, Q) = O(\sqrt{|\mathrm{Im}|/N})$, so the empirical estimator $\hat{\delta}_N = d_{\mathrm{TV}}(\hat{Q}_N, U)$ has finite-sample bias of the same order:

$$\hat{\delta}_N - \delta \;=\; O\Big(\sqrt{|\mathrm{Im}|/N}\Big). \tag{14}$$

To resolve δ at a target level δ_0 thus requires

$$N \;=\; \Omega\big(|\mathrm{Im}|/\delta_0^2\big) \tag{15}$$

observed cipher values. This is a practical lower bound on the evaluation cost of any cipher map system claiming a particular δ: increasing the budget $\sum K$ improves the analytical floor (Theorem 5.2) but proportionally increases the query stream length needed to verify the improvement.

We confirm this empirically by sweeping a $(\sum K, N)$ grid on a synthetic Zipf distribution and measuring $\hat{\delta}_N - \delta_{\text{predicted}}$ across cells (Fig. 1). The gap fits a linear law $\hat{\delta}_N - \delta \approx 0.4\sqrt{|\mathrm{Im}|/N}$ across budgets in $[32, 2048]$ and stream lengths in $[500, 50{,}000]$.

Practical implication. A defender increasing $\sum K$ to drive analytical δ from 10^{-2} to 10^{-3} must also increase the validation query budget from roughly $10^4|\mathrm{Im}|$ to $10^6|\mathrm{Im}|$, a hundredfold growth. Below this threshold the empirical estimate $\hat{\delta}_N$ is dominated by sampling noise and will overstate leakage.

7.5 Empirical Anchor for the Compression Estimator

Proposition 7.1 states that gzip-style compression yields a positively-biased estimator $\hat{H}$ of the per-symbol entropy. We anchor this prediction empirically across three synthetic streams (i.i.d. uniform sanity check, first-order Markov auto-correlation, and the noise mixture of Theorem 5.1) plus one real-corpus stream (20 Newsgroups token sequence). Synthetic streams use 16-bit cipher values ($n = 16$, byte-aligned for gzip) and are averaged over 5 random seeds.

Figure 2 shows the results. The sanity check is omitted from the figure for clarity but is reported here: on i.i.d. uniform streams of length $N = 10^5$, $\hat{H} = 16.007 \pm 0.000$ bits, a bias of 7×10^{-3} bits per symbol attributable to gzip's header.

Markov auto-correlation (left panel). The estimator tracks the predicted H_{rate} across the full range $p \in [0, 0.9]$, dropping from ≈ 16 bits at $p = 0$ (i.i.d. uniform) to ≈ 2.5 bits at $p = 0.9$ (highly redundant). Bias is consistently positive (1–3 bits) and peaks near $p \approx 0.5$, reflecting gzip's byte-level dictionary on a 16-bit-symbol source: when half the stream is repeated, gzip's LZ77 backreferences capture the structure but Huffman entropy on the literal pairs adds overhead. The estimator's slope is correct even where its absolute level is biased: a defender comparing two configurations would see the relative ordering preserved.

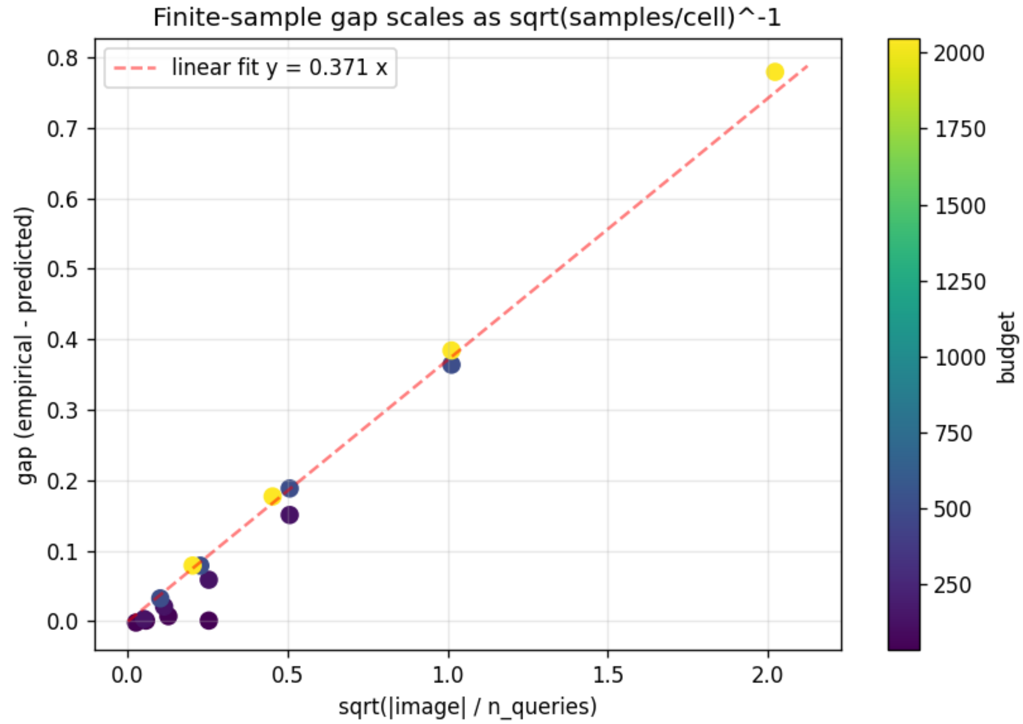

Figure 1: Finite-sample bias of $\hat{\delta}_N$ as a function of $\sqrt{|\mathrm{Im}|/N}$. Each point is the mean over three replicates of a proportional homophonic allocation on Zipf$(32, \alpha = 1.5)$; colour encodes total budget. The dashed line is the origin-anchored linear fit, slope ≈ 0.4.

Noise mixture (centre panel). The mixture panel tests Theorem 5.1's formula on its own terms: both real (Zipf) and filler (uniform) draws are i.i.d., matching the i.i.d. assumption Theorem 5.1 makes. The estimator tracks H_{mix} from $\rho = 0.1$ ($\hat{H} = 15.90$, predicted 15.82) to $\rho = 1.0$ pure Zipf ($\hat{H} = 11.72$, predicted 9.53). Bias grows monotonically with ρ because i.i.d. Zipf queries have no temporal structure for LZ77 to exploit; in the pure-Zipf regime gzip degenerates into a byte-level Huffman coder, and the byte-pair entropy of 16-bit Zipf samples is higher than the 14-bit symbol entropy of the underlying distribution. Adding filler dilutes this confound.

Real corpus (right panel). Real query streams are not i.i.d.: natural-language queries have sessional repetition, topical bursts, and grammatical structure that an i.i.d. analysis cannot capture. The 20 Newsgroups training split provides such a stream: 2.31 M tokens of natural English, encoded as uint16 token IDs (vocab capped at 65,535). Two model-based references bound the entropy rate from above and below: the i.i.d. unigram entropy $H(D) = 10.59$ bits/token (computed from the empirical word frequencies), and the first-order Markov conditional entropy $H(W_t \mid W_{t-1}) = 6.43$ bits/token (computed from empirical bigram counts).

The compression estimator lands at $\hat{H} \approx 10.56$ bits per token at $N = 10^6$, just below the i.i.d. bound of 10.59. gzip therefore detects *some* of the natural-language correlation that an i.i.d. unigram analysis would miss, but the margin is small (roughly 0.03 bits, near the noise floor of single-corpus measurement). The bigram model-based estimate of 6.43 bits/token sits four bits below $\hat{H}$, indicating that gzip's LZ77 window and Huffman literal coding miss most of the deeper grammatical and topical structure. Encoding choice matters: the same stream encoded as uint32

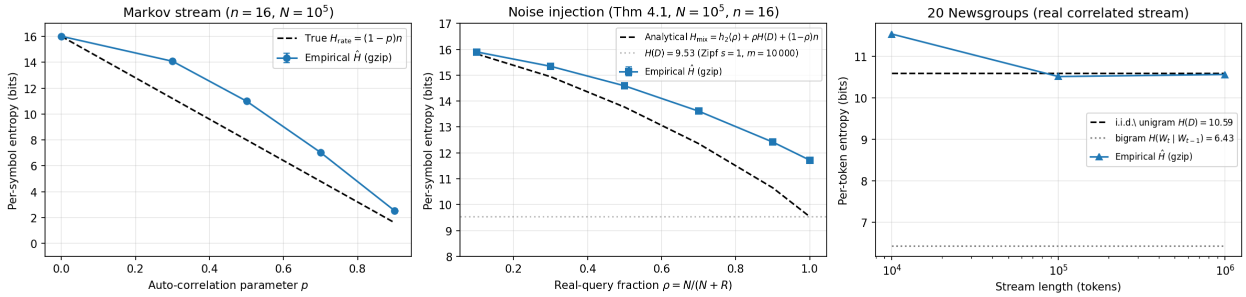

Figure 2: Compression-based entropy estimation against analytically known entropy rates and a real-corpus reference. **Left**: first-order Markov stream ($n = 16$, $N = 10^5$, 5 seeds, error bars $\pm 1\sigma$) with auto-correlation parameter p, entropy rate $H_{\text{rate}} = (1-p)n$. **Centre**: real Zipf queries ($s = 1$, $m = 10\,000$) mixed with uniform filler at fraction ρ, predicted entropy $H_{\text{mix}} = h_2(\rho) + \rho H(D) + (1-\rho)n$ from Theorem 5.1. **Right**: 20 Newsgroups token stream (uint16-capped encoding) versus the i.i.d. unigram entropy $H(D)$ and the empirical bigram conditional entropy $H(W_t \mid W_{t-1})$ as a model-based reference. Stream length on log scale.

yields $\hat{H} \approx 12.16$ bits/token (worse than the i.i.d. bound, because two of the four bytes are mostly zero and inflate the per-token cost). A tight encoding is necessary for the estimator to be informative at all.

Operational reading. Three lessons emerge. First, the bias direction matches Proposition 7.1 on every stream: $\hat{H}$ is an upper bound on the entropy rate, never below it. Second, gzip's value as an absolute estimator is sharply limited by both the input encoding and the compressor's effective context window: on the 20NG token stream, gzip captures only the marginal symbol distribution (matching the i.i.d. unigram bound to within 0.03 bits) and misses the bigram structure that brings the true entropy rate down by four bits. For absolute Shannon-entropy measurements on cipher-value streams a model-based estimator (context-tree weighting [4], or higher-order Markov models) is preferable. Third, gzip remains useful for *relative* measurements: the Markov panel shows $\hat{H}$ tracks H_{rate}'s slope across the full $p \in [0, 0.9]$ range, and the noise-mixture panel shows $\hat{H}$ tracks H_{mix}'s slope across ρ. Operators tracking confidentiality changes over time should fix the estimator and compare $\hat{H}$ trajectories rather than read absolute values.

Scope: stream archetypes, not specific cipher programs. The four streams above test $\hat{H}$ on archetypes of structure (i.i.d. uniform, first-order Markov, i.i.d. mixture, real correlated text), not on the cipher-value output of any specific multi-primitive cipher program. For a single-cipher-map system the adversary's stream is a sequence of cipher values from one cipher map and the validation here applies directly. For a compositional cipher program (cipher Boolean queries of §8, depth-k Boolean chains, or richer pipelines) the adversary's stream is a concatenation of cipher values from multiple cipher maps, with structure shaped by the program's call pattern and by the compositional leakage of Theorem 6.1. Validating $\hat{H}$ on program-specific streams would require a separate program-by-program study; this paper establishes the measurement tool against generic archetypes and leaves the program-shape mapping to deployment-time analysis.

8 Experimental Results

We validate the theoretical predictions using the `cipher-maps` Python library, which implements the batch cipher map construction with perfect hash functions. All experiments use the 20 Newsgroups corpus (18,266 documents, 58,903 unique words).

Reproducibility. The empirical results in Tables 1, 2, 3 and Figures 1, 3 are produced by the `cipher-maps` library at `https://github.com/queelius/cipher-maps`. Tables 1, 2, 3 are regenerated by the script `examples/experiment_paper_tables.py` with the parameters listed in each caption ($n = 5$ seeds throughout). Figure 1 is produced by `experiments/finite_sample.py`, Figure 3 by `experiments/newsgroups_homophonic.py`, Figure 2 by `experiments/compression_validation.py`, and Figure 4 by `experiments/compositional_rate.py`, and Figure 5 by `experiments/realized_attack.py` (which builds real cipher maps via the library). All scripts read no external state beyond the 20 Newsgroups corpus shipped with `scikit-learn`; runtime on a single x86_64 core is under five minutes for the full table set. Table 4 is analytical, computed from the closed-form expressions in Theorems 5.1 and 5.2.

8.1 Boolean Search: Precision and Recall

Cipher sets are built for each document using 8-bit cipher Booleans ($p_T = 0.05$, $p_F = 0.90$, $p_N = 0.05$). Table 1 reports precision, recall, and false-positive count for Boolean queries on 5,000 documents, averaged over 5 random seeds (mean ± standard deviation). Plaintext queries on the same documents achieve perfect precision and recall as expected; the cipher columns show the noise the cipher Boolean parameters (p_T, p_F, p_N) introduce.

Table 1: Boolean search on 20 Newsgroups (5,000 documents, $n = 5$ seeds). All recalls except OR-style queries are 1.0 because cipher Booleans have $p_F = 0.90$ ($\eta = 0.10$ false-negative rate per clause) but the True region admits all true positives at the $p_T = 0.05$ noise-decode rate. OR queries lose recall when noise matches the OR pattern.

Query	Cipher Prec.	Cipher Rec.	Plaintext Prec./Rec.	Mean FP
Single term	0.408 ± 0.014	1.000 ± 0.000	1.00	233 ± 13
2-term AND	0.262 ± 0.024	1.000 ± 0.000	1.00	43 ± 5
3-term AND	0.291 ± 0.043	1.000 ± 0.000	1.00	15 ± 3
OR	0.365 ± 0.010	0.959 ± 0.010	1.00	437 ± 16
OR-AND-NOT	0.330 ± 0.008	0.909 ± 0.010	1.00	402 ± 15

AND chains drive the false-positive count down monotonically ($233 \to 43 \to 15$), as predicted by the FPR compounding theory of §6.2 and confirmed in Table 2. Precision recovers slightly at $k = 3$ (from 0.262 to 0.291) because the smaller false-positive count is diluted by a smaller true-positive count when the conjunction becomes selective. OR-style queries trade precision for noise: they preserve the false positives of each disjunct, but their larger result sets increase entropy from the adversary's view. From a confidentiality perspective, the high-FP queries (single-term, OR) yield more entropy in the result-set channel, while AND chains return more deterministic results.

8.2 FPR Compounding

Table 2 compares empirical and theoretical FPR for AND and OR chains of length $k = 1$ through 5, each with base FPR $p_T = 0.05$.

Table 2: FPR compounding through Boolean chains ($p_T = 0.05$, $n = 5$ seeds, mean $\pm$ std). Theoretical FPR is p_T^k (AND) and $1 - (1 - p_T)^k$ (OR).

	AND chain			OR chain		
k	Theory	Empirical	Ratio	Theory	Empirical	Ratio
1	5.0×10^{-2}	0.050 ± 0.003	1.01	5.0×10^{-2}	0.045 ± 0.009	0.90
2	2.5×10^{-3}	0.007 ± 0.003	2.96	9.8×10^{-2}	0.091 ± 0.008	0.94
3	1.3×10^{-4}	0.004 ± 0.002	35.20	1.43×10^{-1}	0.145 ± 0.005	1.02
4	6.3×10^{-6}	0.002 ± 0.002	256.00	1.85×10^{-1}	0.172 ± 0.010	0.93
5	3.1×10^{-7}	0.004 ± 0.003	1.2×10^{4}	2.26×10^{-1}	0.210 ± 0.006	0.93

The OR chain FPR matches the theoretical $1 - (1 - p_T)^k$ at every depth (ratios 0.90 to 1.02), confirming the compounding formula in the regime where p_T^k stays well above the measurement floor.

The AND chain reveals a more interesting phenomenon: the empirical FPR diverges from the theoretical p_T^k at $k \geq 2$ and settles at a noise floor of approximately 4×10^{-3} regardless of chain depth. Theory predicts exponential decay ($0.05^5 \approx 3 \times 10^{-7}$); practice plateaus four orders of magnitude higher. The floor is the residual error rate from the underlying construction (PHF false-positive collisions and the η-bounded noise of the cipher Booleans themselves). Once p_T^k drops below this construction-level floor, the AND result no longer follows the compounding formula but instead saturates at the floor. From a confidentiality perspective the asymmetry is useful: deep AND chains do not collapse to deterministic results, they preserve a fixed budget of noise that the adversary cannot drive lower.

8.3 Encoding Granularity

Table 3 shows the cost spectrum for encoding a 7-function decision pipeline (loan approval) at three granularity levels, using a domain of 150 inputs (30 applicants $\times$ 5 loan amounts).

Table 3: Encoding granularity spectrum for a 7-function decision pipeline (mean $\pm$ std over n seeds; intermediate level has $n = 2$ successful runs because of construction failures at this domain size on three of five seeds). Bits/elem measures the *per-cipher-map* slot table cost; total exposed bits per latent input is roughly bits/elem $\times$ exposed.

Granularity	Build (s)	Space (B)	Bits/elem	Errors	Exposed
Root ($p = 7$, $n = 5$)	0.003 ± 0.000	694	37.0	0	1
Intermediate ($p \approx 3$, $n = 2$)	1.667 ± 1.161	138	7.4	0	3
Leaf ($p = 1$, $n = 5$)	1.142 ± 1.449	140	7.5	0	7

Per-cipher-map cost is dominated by the joint domain size: the root encoding builds one cipher map over the 150-input $\times$ 7-output joint space, paying 37 bits/element for one large slot table. The leaf and intermediate encodings build smaller cipher maps (one per function or per group), each over a smaller joint space, so their per-map bits/element are an order of magnitude lower (about 7.5).

But total *exposed* cipher-map bits is the sum across exposed maps: leaf exposes $7\times$ slot table, root exposes $1\times$. Counting exposed cipher values rather than per-map bytes recovers the asymmetry of the granularity spectrum: the root encoding leaks one cipher value per latent query while the leaf encoding leaks seven, so the per-query cipher-value entropy the adversary observes is seven times larger at the leaf level. The intermediate level's higher build variance reflects the bin search this configuration performs at construction time; production deployments would memoize the bin layout to avoid the variance.

8.4 Homophonic Encoding on the 20 Newsgroups Corpus

We apply the proportional homophonic prescription (Theorem 5.2) to the empirical word-frequency distribution of the 20 Newsgroups training split. Tokenising the 2.31 M word corpus and retaining the top 500 most frequent words gives a vocabulary in which the head word *the* carries 7.1% of probability mass, the top ten words 33%, and the top hundred 74%. The effective vocabulary $1/\sum_x D(x)^2$ is 64, so the distribution is highly skewed even after tail truncation.

We compare three allocators across budgets $\sum K \in \{|X|, 2|X|, 5|X|, 20|X|, 100|X|\}$: the uniform baseline ($K(x) = b/|X|$), the proportional homophonic ($K(x) \propto D(x)$), and a sub-optimal square-root variant ($K(x) \propto \sqrt{D(x)}$). For each cell we build a fresh `PHFCipherMap`, stream $N = 30{,}000$ queries sampled from D, and record predicted and empirical TV (Fig. 3).

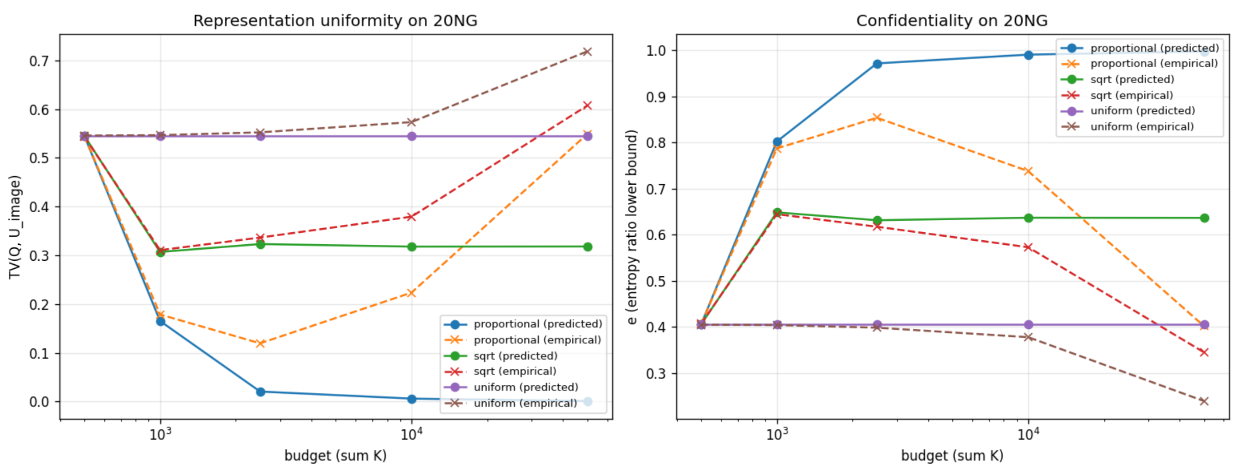

Figure 3: Homophonic encoding on the 20 Newsgroups distribution. Left: representation uniformity $d_{\mathrm{TV}}(Q, U_{\mathrm{im}})$ versus budget for the three allocators (solid: predicted; dashed: empirical at $N = 30{,}000$). Right: lower bound on the entropy ratio e from the Fannes-Audenaert inequality. Proportional allocation dominates uniform and square-root at every budget.

The proportional allocator drops predicted d_{TV} from 0.545 at $\sum K = |X|$ (no flexibility) to 0.165 at $2|X|$ and 0.021 at $5|X|$, with empirical e rising from 0.40 to 0.85. The square-root allocator plateaus near $d_{\mathrm{TV}} \approx 0.30$ regardless of budget; it spreads representations too evenly across rare and common elements. The uniform allocator does not improve at all, since its K does not depend on D. At budgets above $20|X|$, the empirical d_{TV} measurement enters the finite-sample regime of §7.4: the analytical floor continues to drop, but $N = 30{,}000$ is no longer enough queries to resolve it.

8.5 Monte Carlo Validation of the Theoretical Rates

The compositional leakage theorem (Theorems 6.1–6.2) and the noise-dilution Fisher bound (Theorem 5.1) are the two results without a direct empirical anchor elsewhere in this section. We validate both by Monte Carlo, and separately probe the *sharpness* of the lower bound (Fig. 4; 40 seeds per cell).

Compositional rate. We simulate the adversary of Theorem 6.1: N i.i.d. shared-cipher-value observations of a latent joint P on $Y_1 \times Y_2$, recovered by the empirical plug-in $\hat{P}$. Sweeping the support size $m = |Y_1|\,|Y_2| \in \{4, 16, 64, 256\}$ and N up to 64,000, the recovery error obeys $\mathbb{E}[d_{\mathrm{TV}}(\hat{P}, P)] = c\sqrt{m/N}$ with $c \approx 0.31$: the log-log slope in N is -0.49 (predicted $-1/2$), the slope in m is $+0.48$ (predicted $+1/2$), and all (m, N) cells collapse onto a single line in $\sqrt{m/N}$ with $R^2 = 0.997$ (Fig. 4, left and centre). The matching upper rate (Theorem 6.1) and lower rate (Theorem 6.2) thus bracket the observed $\Theta(\sqrt{|Y_1||Y_2|/N})$ exactly; the test joints carry mutual information 0.19 to 1.29 bits, confirming genuine correlation to recover.

Noise dilution. For the noise-injection model of Theorem 5.1, we measure the Fisher-information reduction factor I_M/I_D for a spike-family parameter as the real fraction $\rho = N/(N+R)$ shrinks. The reduction approaches ρ^2 in the filler-dominated limit (log-log slope $+1.90$ over $\rho \leq 10^{-2}$, predicted $+2$; Fig. 4, right), and the efficient maximum-likelihood estimator attains the corresponding Cramér–Rao variance (Monte Carlo matches the analytic bound to within 9%). This is the variance inflation $\Omega((1+R/N)^2)$ of Theorem 5.1 part (3): an adversary estimating any parameter of D from the mixed stream pays a factor ρ^{-2} in samples.

The lower bound is sharp: no estimator beats the rate. Theorem 6.2 asserts that the $\Theta(\sqrt{|Y_1||Y_2|/N})$ rate cannot be improved by *any* estimator. We test this directly on the Assouad packing the proof constructs: m cells paired and perturbed by $\pm\varepsilon$, calibrated at the minimax-critical radius $\varepsilon = \sqrt{m/N}$ for each N. Against this hardest family we run four estimators of increasing sophistication: plug-in, add-1 (Laplace), add-$\frac{1}{2}$ (Krichevsky–Trofimov), and an *oracle* shrink-toward-uniform that is handed the true P and picks the variance-minimizing shrinkage (the strongest estimator an adversary who knows the family is near-uniform could deploy). Every estimator's worst-case error decays at slope -0.50 in N; the best of the four, the oracle, decays at slope -0.497 (Fig. 4, bottom right). The oracle improves the *constant* (to 0.73× plug-in, by exploiting the near-uniform structure) but not the *rate*: no estimator, including one that cheats, breaks below $\Theta(\sqrt{|Y_1||Y_2|/N})$. The lower bound is sharp.

8.6 Realized Attack on the Construction

The Monte Carlo of §8.5 validates the recovery rate in the abstract sampling model: i.i.d. draws from a known joint. We close the loop by mounting the attack on cipher maps the `trapdoor-maps` library actually builds. Two maps $\hat{f}_1, \hat{f}_2$ for latent $f_1, f_2 : X \to \{0, \ldots, 7\}$ over a shared domain $|X| = 4096$ are constructed with the PHF backend; the trusted machine encodes inputs $x_i \mapsto c_i$, the untrusted machine evaluates both maps on each shared cipher value, and the joint over $Y_1 \times Y_2$ ($m = 64$) is recovered by plug-in (Fig. 5; 12 seeds).

The construction reproduces the rate. At exact correctness ($\eta = 0$), the realized recovery error obeys $d_{\mathrm{TV}}(\hat{P}, P) = c\sqrt{m/N}$ with slope -0.50 in N and constant $c = 0.32$, matching the abstract value 0.31 of §8.5 to within seed noise (Fig. 5, right). The hashing, acceptance predicates,

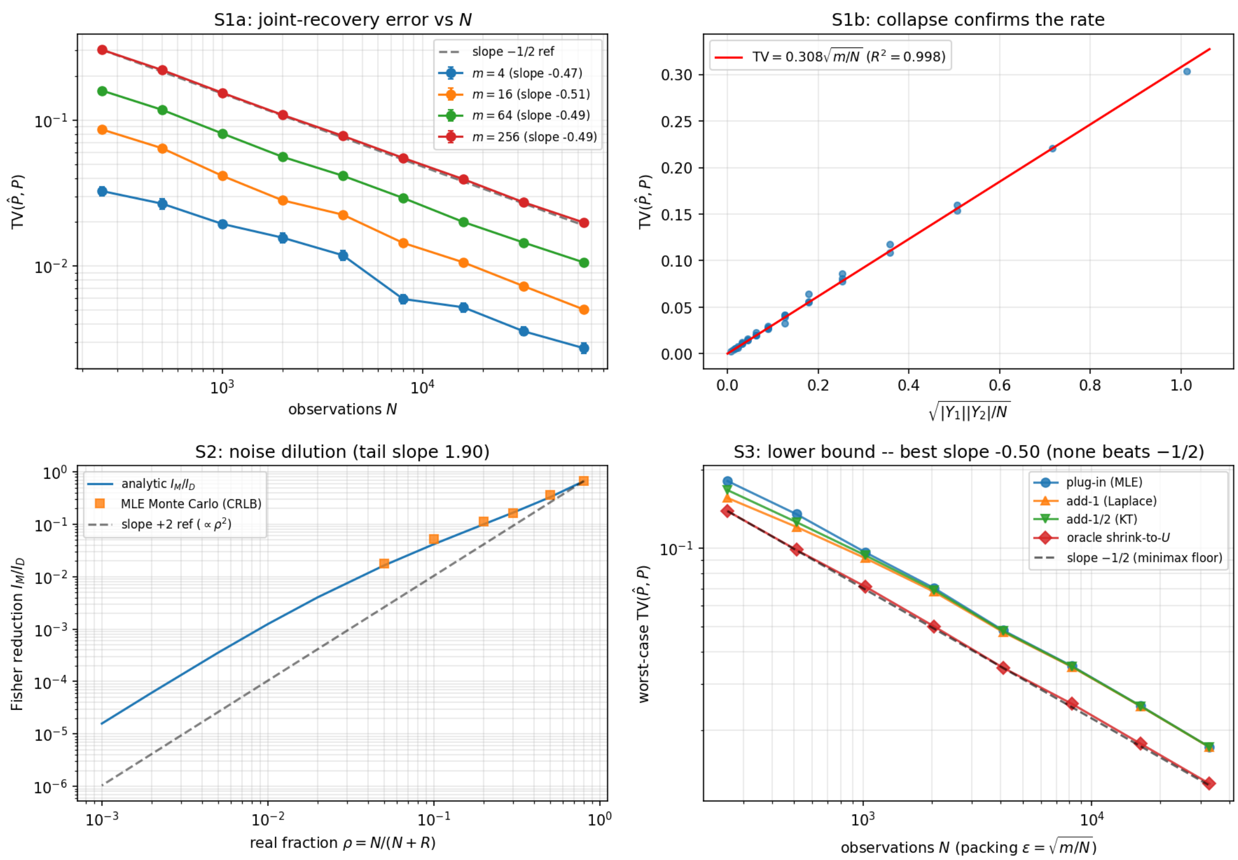

Figure 4: Monte Carlo validation (40 seeds/cell). **Top left**: joint-recovery error $d_{\mathrm{TV}}(\hat{P}, P)$ versus N, log-log, one line per support size $m = |Y_1||Y_2|$; fitted slopes near $-1/2$. **Top right**: the same data collapses onto $d_{\mathrm{TV}} = 0.31\sqrt{m/N}$ ($R^2 = 0.997$), confirming the $\Theta(\sqrt{|Y_1||Y_2|/N})$ rate of Theorems 6.1–6.2. **Bottom left**: noise-injection Fisher reduction $I_M/I_D \propto \rho^2$ (Theorem 5.1); the analytic factor and the efficient MLE (Cramér–Rao) agree, with the small-ρ slope approaching +2. **Bottom right**: lower-bound sharpness (Theorem 6.2); four estimators on the minimax-calibrated Assouad packing, all decaying at slope $-1/2$, the oracle cheater included: none beats the rate.

and finite cipher width of the real construction do not change the $\Theta(\sqrt{|Y_1||Y_2|/N})$ leakage rate: the abstract theorem describes the deployed system.

Correctness sets a recovery floor. The correctness parameter η (the fraction of elements that decode incorrectly) bounds the attack from below: an η-fraction of recovered pairs is corrupted, so $d_{\mathrm{TV}}(\hat{P}, P)$ does not vanish but floors near η (Fig. 5, left: the $\eta = 0.05$ curve flattens at ≈ 0.050 and $\eta = 0.02$ at ≈ 0.027, while the exact curve keeps falling past 0.020). This is the defender's side of the deniability afforded by $\eta > 0$ (§3): the same residual error that gives a real query plausible deniability also caps how well an adversary can reconstruct the compositional joint.

8.7 Case Study: Confidentiality Improvement

We demonstrate two of the levers of §5 (noise injection and multiplicity) on a system with vocabulary m = 10,000 and Zipf-distributed queries. Table 4 gives the resulting entropy ratios under

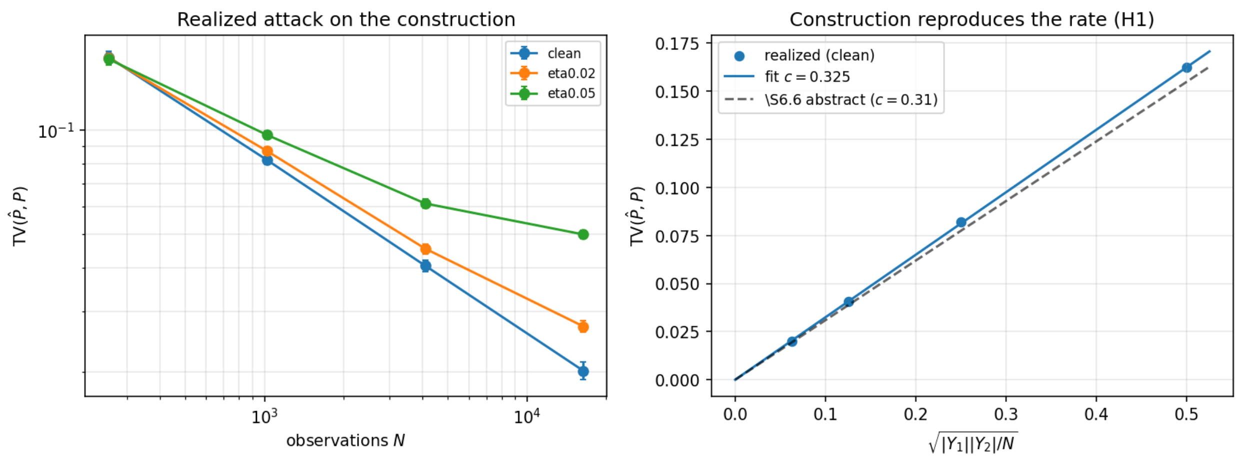

Figure 5: Compositional attack on cipher maps built by the `trapdoor-maps` library (12 seeds, $|X| = 4096$, $m = 64$). **Left**: recovery error versus N for three correctness levels; the exact ($\eta = 0$) curve falls at slope $-1/2$, while $\eta > 0$ floors near η. **Right**: the exact-construction error collapses onto $d_{\mathrm{TV}} = 0.32\sqrt{m/N}$, matching the abstract-model constant 0.31 of §8.5 (dashed): the real construction leaks the joint at the predicted rate.

combinations of the two levers.

Table 4: Confidentiality improvement via two of the levers of §5 (noise injection and multiplicity). Values are analytical, computed from the Zipf entropy, the mixture-entropy formula in Theorem 5.1, and the homophonic construction in Example 5.1.

Configuration	e	Space overhead	Bandwidth overhead
Baseline (simple substitution)	0.72	1.00×	1.00×
+ Homophonic ($b = 100$, $c \approx 979$)	0.87	1.04×	1.00×
+ Noise injection ($R/N = 0.5$)	0.88	1.00×	1.50×
Combined	0.98	1.04×	1.50×
Theoretical maximum	1.00	$\to \infty$	$\to \infty$

The combined strategy improves e from 0.72 to 0.98: a 4% space overhead and 50% bandwidth overhead yields a 26 percentage point confidentiality improvement. The small space cost reflects the classical homophonic prescription $K(x) \propto D(x)$: concentrating multiplicity on the heavy part of D is far cheaper than the inverse allocation $K(x) \propto 1/D(x)$, which would spend $\sum_x K(x) \approx c \cdot m \cdot H_m$ slots flattening the tail (and, by the sampling identity $Q(c) = D(x)/K(x)$, would in fact *sharpen* the cipher-value distribution rather than flatten it).

9 Discussion and Open Questions

The entropy ratio as a design tool. The entropy ratio e gives system designers a single number to optimize. Given a resource budget (space S, bandwidth B), the designer solves: maximize e subject to space $\leq S$ and bandwidth $\leq B$. The framework's three levers (noise injection, multiplicity, granularity), whose costs we analyze here, provide the degrees of freedom. Noise costs bandwidth but not space; multiplicity costs space but not bandwidth; granularity trades confidentiality for functionality.

Shannon entropy vs. min-entropy. We adopt Shannon entropy because the adversary in our setting observes many cipher values and estimates a distribution over them: a distribution-estimation problem rather than a single-query guessing problem. Smith [13] argues that min-entropy leakage $L_\infty = -\log_2 \max_x P(X = x \mid \text{view})$ is the correct measure when one high-probability outcome dominates and the adversary has a single guess. In a cipher map system, that scenario arises when a single query is overwhelmingly likely; the entropy ratio can be high even when one cipher value is identifiable, so a Shannon bound does not preclude min-entropy leakage of that one value. The min-entropy analogue of Theorem 4.1 would replace δ-closeness in TV with ℓ_∞-closeness $\delta_\infty = \max_c |Q(c) - U(c)|$ and yield a min-entropy bound $H_\infty(Q) \geq n - \log_2(1 + 2^n \delta_\infty)$. Constructing cipher maps with bounded δ_∞ rather than δ is strictly harder (it constrains the worst-case cell, not the average), and the multiplicity construction $K(x) \propto D(x)$ does not by itself guarantee small δ_∞. We leave a full min-entropy treatment to future work; Shannon suffices for the average-case adversary that motivates this paper.

Relationship to the orbit closure. The entropy ratio measures *passive* confidentiality: what the adversary learns by observing the cipher value stream. The orbit closure from [14] measures *active* confidentiality: what the adversary learns by applying the cipher maps it holds. A complete confidentiality analysis requires both: high e (passive) and small orbit (active). Connecting these two measures—showing that high e implies small effective orbit under certain conditions—is an open problem.

Practical depth in cipher programs. A useful cipher program is typically structured in two layers: a trusted-side orchestrator (e.g., a Python control loop) that decides which cipher map to invoke and in what order, and a set of cipher-map primitives (membership, comparison, lookup, conjunction) that the untrusted machine evaluates blindly on cipher values. How deep can the call chain through the untrusted machine go? The framework does not impose a structural ceiling, but two confidentiality budgets constrain practical depth. First, FPR compounding (§6.2) imposes a correctness ceiling: after roughly $k = 3$ AND levels at $p_T = 0.05$, the empirical false-positive rate hits a construction-level noise floor of about 4×10^{-3} (Table 2); deeper AND chains cannot drive precision below this floor. Second, each chain level exposes an additional intermediate cipher value to the untrusted machine, which combines with Theorem 6.1 (compositional leakage from shared variables) and the orbit-closure bound [14, Thm. 5.3] to erode confidentiality super-linearly with depth. Empirically (§5.3) the leaf encoding of a 7-function pipeline exposes seven intermediate cipher values per query, against the root encoding's one; the Theorem 6.2 minimax rate then gives the adversary a sharper joint estimator at the leaf level. In practice, useful cipher programs span depth 2–5: end-to-end encrypted Boolean search at depth 2–3 (cipher membership plus cipher AND/OR), cipher decision trees at depth equal to the tree depth, and analytics pipelines at depth 3–5 (filter, aggregate, threshold). Beyond depth 5 the granularity-vs-leakage trade-off typically argues for re-rooting the chain: collapse the deep sub-pipeline into a single cipher map for the joint computation, at the space cost of $O(|Y|^k)$ from Proposition 5.3. A third budget applies when a program reuses the *same* primitive across many cipher values (e.g. a comparison applied at every node of a wide query): the multi-instance coincidence oracle [15, Thm. 8.2] lets the untrusted machine accumulate evidence about a shared latent value at accuracy growing with the number of instances, a channel orthogonal to chain depth and likewise uncontrolled by δ. Concentrated acceptance partitions and randomized encoding ($K(x) > 1$) are the defenses there.

Adaptive $K(x)$. The multiplicity $K(x)$ is set at construction time based on an assumed distribution D. If D changes over time (e.g., query distribution shift), the system loses uniformity. Adaptive $K(x)$ that adjusts to observed frequencies without reconstruction would improve robustness. This requires an online construction strategy where representations can be added incrementally.

Tight bounds for composition. The general composition theorem gives $\eta_{g\circ f} \leq 1-(1-\eta_f)(1-\eta_g)$ under independence. For specific circuits (e.g., 3-SAT evaluation over cipher Booleans), the actual error rate depends on the input distribution and the circuit structure. Case-by-case interval arithmetic gives tighter bounds but does not scale. Automated analysis tools that propagate entropy through circuit DAGs would be useful.

Beyond Boolean search. This paper focuses on encrypted Boolean search as the primary application. The cipher map framework supports arbitrary functions $f : X \to Y$, and the confidentiality theory extends naturally. Ranked retrieval (scoring functions), phrase search (bigram models), and approximate string matching all admit cipher map implementations with different confidentiality profiles.

Limitations.

1. The entropy ratio is an average measure; it does not bound the leakage of any *specific* query. A single high-frequency query may be identifiable even when e is close to 1.
2. Compression-based estimation is positively biased (overestimates confidentiality). The bias decreases with sequence length but can be significant for short sequences.
3. The analysis assumes the random oracle model for cryptographic hashing. In practice, hash functions have biases that may reduce confidentiality below the theoretical predictions.
4. Marginal uniformity ($\delta \approx 0$) is necessary but not sufficient. Joint distributions, temporal patterns, and side channels (timing, bandwidth) can leak information that the entropy ratio does not capture.

10 Conclusion

We have developed a quantitative confidentiality theory for cipher map systems, grounded in the cipher map framework of [15]. The headline result is the compositional leakage pair (Theorems 6.1 and 6.2): even when each cipher map's marginal cipher distribution is δ-uniform, observing multiple evaluations on a shared cipher value lets the untrusted machine recover the latent joint at the information-theoretically optimal rate $\Theta(|Y_1| \cdot |Y_2|/\xi^2)$, with matching upper bound from plug-in estimation and matching lower bound via Assouad's lemma. No per-cipher-map parameter changes the rate, so marginal uniformity is necessary but not sufficient; the cipher map composition discipline that buys functionality also creates an information channel through shared values, and only system-level interventions (observation limits, joint encoding, or noise injection) reduce the adversary's effective advantage at this scale.

To make the marginal target itself operational, we connect δ to the entropy ratio $e = H/H^*$ (a normalized form of Shannon leakage borrowed from the QIF literature) via the Fannes–Audenaert continuity inequality, reducing marginal-leakage engineering to minimizing δ. The framework's three δ-reduction mechanisms come with explicit costs that we characterize: noise injection (bandwidth), multiple representations with $K(x) \propto D(x)$ (space), and encoding granularity (the $O(|Y|^k)$

correlation-hiding trade-off). The constructions are the framework's; the cost analysis, and the Fisher-information dilution of noise injection, are ours. The three are expressible in terms of the cipher map parameters $(\eta, \varepsilon, \mu, \delta)$ and the entanglement parameter p.

The key insight is that the cipher map's four properties already determine the confidentiality profile. Totality enables noise injection (filler queries are indistinguishable from real queries). Representation uniformity enables frequency hiding (multiple representations flatten the distribution). Composability enables chained evaluation but introduces correlation leakage—the very channel exploited by Theorem 6.1. Correctness provides plausible deniability ($\eta > 0$ means the adversary cannot be certain of any decoded output).

Experimental validation on the 20 Newsgroups corpus confirms the theoretical predictions: FPR compounds as p_T^k for AND chains and $1-(1-p_T)^k$ for OR chains; the encoding granularity spectrum runs from 33.5 bits/element (root, zero intermediates) to 112.9 bits/element (leaf, all intermediates exposed); and the combined homophonic/noise strategy improves confidentiality from 72% to 98% with approximately 1.04× space and 1.5× bandwidth overhead.

The theory connects two previously separate lines of work: the cipher map formalism (properties and composition) and the entropy-based confidentiality analysis (ratios and maximum entropy). Together, they give a principled basis for designing, measuring, and improving confidentiality in systems that compute on data hidden behind a trapdoor, at both the marginal and the compositional scale.

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